FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

PART TEST – III

PAPER-1

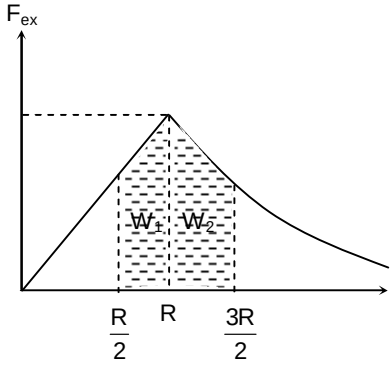
Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**A, C**	21.	**A, B, C, D**	41.	**A, D**
2.	**A, D**	22.	**A, B**	42.	**A, B, D**
3.	**A, B, C**	23.	**A, B, D**	43.	**A, B, D**
4.	**B, C**	24.	**A, C, D**	44.	**A, B, C**
5.	**B, D**	25.	**A, B, C, D**	45.	**A, B, C**
6.	**A, B, C, D**	26.	**A, B, D**	46.	**A, B**
7.	**A, B**	27.	**A, B, C, D**	47.	**A, C, D**
8.	**A, B, C**	28.	**A, D**	48.	**A, C**
9.	**A**	29.	**A, B, C, D**	49.	**A, B, C, D**
10.	**A**	30.	**A, B, C**	50.	**A, B, C**
11.	**(A) → (p, r), (B) → (s), (C) → (q, s), (D) → (p, r)**	31.	**(A) → (t), (B) → (r, s), (C) → (p, q), (D) → (p, q, s)**	51.	**(A) → (q, r, t), (B) → (q, r, t), (C) → (p, q), (D) → (p, r, s, t)**
12.	**(A) → (q, s), (B) → (q, s), (C) → (p, r), (D) → (q, s)**	32.	**(A) → (p, q, r), (B) → (q, s), (C) → (q, r, s, t), (D) → (p, q, r, t)**	52.	**(A) → (p, r, t), (B) → (s, t), (C) → (s), (D) → (p, r, t)**
13.	**4**	33.	**3**	53.	**1**
14.	**8**	34.	**8**	54.	**1**
15.	**2**	35.	**3**	55.	**8**
16.	**5**	36.	**5 m^3**	56.	**1**
17.	**6**	37.	**3**	57.	**1**
18.	**2**	38.	**2**	58.	**1**
19.	**8**	39.	**6**	59.	**4**
20.	**1**	40.	**5**	60.	**9**

Physics

PART – I

SECTION – A

1. $$F_{ex}=\begin{cases}\dfrac{GMm}{R^3}\vec{r} & \text{if } 0\le r\le R\\ \dfrac{GMm}{r^3}\vec{r} & \text{if } R\le r<\infty\end{cases}$$



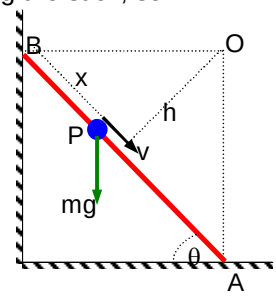
$\Rightarrow \quad W_1 > W_2$

$$W_1=\int_{R/2}^{R}\frac{4}{3}\pi G\rho m x\,dx=\frac{3GMm}{8R}$$

$$W_2=\int_{R}^{3R/2}\frac{GMm}{x^2}dx=\frac{GMm}{3R}$$

$$\Rightarrow \quad \frac{W_1}{W_2}=\frac{9}{8}\Rightarrow\left|\frac{W_1+W_2}{W_1-W_2}\right|=17$$

2. Consider the bug at point P at any time t, moving with speed v along the stick, so $L=mvh=mv\ell\sin\theta\cos\theta\Rightarrow$ Angular momentum of bug about point O



$$\Rightarrow \tau=\frac{dL}{dt}=\left(m\ell\sin\theta\cos\theta\right)\frac{dv}{dt}$$

$$\Rightarrow mg\left(\ell-x\right)\cos\theta=\left(m\ell\sin\theta\cos\theta\right)\frac{dv}{dt}$$

$$\Rightarrow \frac{dv}{dt}=\frac{g}{\sin\theta}\left(1-\frac{x}{\ell}\right)$$

$\Rightarrow \dfrac{dv}{dt}+\dfrac{g}{\ell\sin\theta}x=\dfrac{g}{\sin\theta}\Rightarrow$ Equation of SHM with mean position at point A

$$\Rightarrow T=2\pi\sqrt{\frac{\ell\sin\theta}{g}}$$

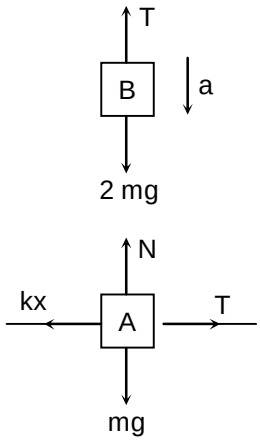
Time required = $\dfrac{T}{4}=\dfrac{\pi}{2}\sqrt{\dfrac{\ell\sin\theta}{g}}$

3. Basic concept of Intensity of light

4. Consider the system at any time t when each block is moving with speed v and spring has an extension x.

$\Rightarrow \quad 2mg - T = 2ma$ …(1)

$\Rightarrow \quad T - kx = ma$ …(2)



From equation (1) & (2), we can write

$$\frac{2g}{3} - \frac{k}{3m}x = a \Rightarrow \frac{d^2x}{dt^2} + \frac{kx}{3m} - \frac{2g}{3} = 0$$

$$\Rightarrow \quad \frac{d^2x}{dt^2} + \frac{k}{3m}\left[x - \frac{2mg}{k}\right] = 0 \Rightarrow \omega = \sqrt{\frac{k}{3m}}$$

For equilibrium $\frac{d^2x}{dt^2} = 0 \Rightarrow x_0 = \frac{2mg}{k}$

Amplitude of motion = $x_0 - 0 = \frac{2mg}{k}$

$$a_{max} = \omega^2 A = \frac{k}{3m} \times \frac{2mg}{k} = \frac{2g}{3}$$

$$v_{max} = \omega A = \frac{2mg}{k}\sqrt{\frac{k}{3m}} = 2g\sqrt{\frac{m}{3k}}$$

Second Method:

Using **COME**, we can write

$$2mgx = \frac{3mv^2}{2} + \frac{kx^2}{2}$$

For extreme position, v = 0 $\Rightarrow$ x = 0, $\frac{4mg}{k}$

A $\Rightarrow$ Amplitude of motion = $\frac{2mg}{k}$

For maximum velocity, $\frac{d^2x}{dt^2} = 0 \Rightarrow x = \frac{2mg}{k} \Rightarrow$ Equilibrium position

$$\Rightarrow \quad 2mg \times \frac{2mg}{k} = \frac{3mv_{max}^2}{2} + k\frac{4m^2g^2}{2k}$$

$$\Rightarrow \quad \frac{4m^2g^2}{2k} = \frac{3mv_{max}^2}{2} \Rightarrow v_{max}^2 = 4g^2\left(\frac{m}{3k}\right)$$

$$\Rightarrow \quad v_{max} = 2g\sqrt{\frac{m}{3k}}$$

$$a_{max} = \left|\frac{2g}{3} - \frac{k}{3m} \times \frac{4mg}{k}\right| = \frac{2g}{3}$$

5. The energy released in α-decay is given by

$$Q = (M_U - M_{Th} - M_{He})c^2$$

Substituting the atomic masses as given in the data, we find

$$Q = (238.05079 - 234.04363 - 4.00260)u \times c^2 = (0.00456\ u)c^2 = (0.00456u)c^2$$

$$= (0.00456u)(931.5\ MeV/u) = 4.25\ MeV.$$

If $^{238}_{92}U$ spontaneously emits a proton, the decay process would be $^{238}_{92}U \rightarrow ^{237}_{91}Pa + ^{1}_{1}H$

The Q for this process to happen is

$$= (M_U - M_{Pa} - M_H)c^2 = (238.05079 - 237.05121 - 1.00783)u \times c^2 = (-0.00825u)c^2$$

$$= -(0.00825\ u)(931.5\ MeV/u) \qquad = -7.68\ MeV$$

Thus, the Q of the process is negative and therefore it cannot proceed spontaneously. We will have to supply an energy of 7.68 MeV to a $^{238}_{92}U$ nucleus to make it emit a proton.

6. $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{50 \times 1 \times 9.8}{0.1}} = 70 m/s$.

$\frac{\lambda}{4} = 1 \Rightarrow \lambda = 4m \Rightarrow f = \frac{v}{\lambda} = \frac{70}{4} = 17.5 Hz$.

$v_{pmax} = \omega A \Rightarrow A = \frac{v_{pmax}}{2\pi f} = \frac{15}{2 \times 3.14 \times 17.5} = 0.136 m = 13.6 \times 10^{-2} m$

7. (C) $\frac{M}{M_0} = \frac{Mc^2}{M_0c^2} = \frac{K + M_0c^2}{M_0c^2} = 1 + \frac{10^{-9}}{1.67 \times 10^{-27} \times 9 \times 10^{20}} \approx 7.68$

(D) $Mc^2 = M_0c^2 + K = 2m_0c^2$

$\Rightarrow$ $M = 2 M_0$

$\Rightarrow$ $\frac{M_0}{\sqrt{1 - \frac{v^2}{c^2}}} = 2M_0$

$\Rightarrow$ $1 = 4 - \frac{4v^2}{c^2}$

$\Rightarrow$ $v^2 = \frac{3c^2}{4} \Rightarrow v = \frac{\sqrt{3}}{2} c = 0.866 c$

8. $\frac{m\lambda}{2} = 40$

$\left(n + \frac{1}{2}\right)\frac{\lambda}{2} = 30$

$\frac{(2n+1)}{m} = \frac{6}{4} \Rightarrow m = \frac{2(2n+1)}{3}$

n = 1 m = 2 λ = 40 cm

n = 4 m = 6 $v = f\lambda \Rightarrow f/40 = f_0$

n = 7 m = 10

λ = 8 second overtone

$f = v/8 = 5f_0$

n = 10 m = 14

$\lambda = \frac{40}{7} \Rightarrow f_3 = \frac{7v}{40}$.

9. Let x be the wave length for K_α-line

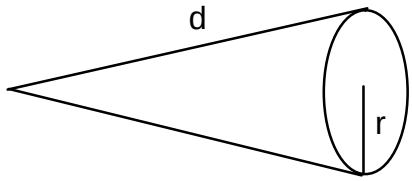
$\frac{x - \frac{hc}{eV_2}}{x - \frac{hc}{eV_1}} = 3$

$x = \frac{4}{3R(Z-1)^2}$

$\Rightarrow$ Z = 29

10. $\frac{dN}{dt} = \frac{P}{(hc/\lambda)} = 5 \times 10^{19}$ / s

$\frac{hc}{\lambda} = 2.49\,eV > W_0$: Yes.



SECTION – B

11. Use $K_{max} = h\nu - \phi$

12. Use concept of combination of lenses.

SECTION – C

13. OE = x

$FG = \sqrt{2}a + 2y$

From the property of rhombus, we can write

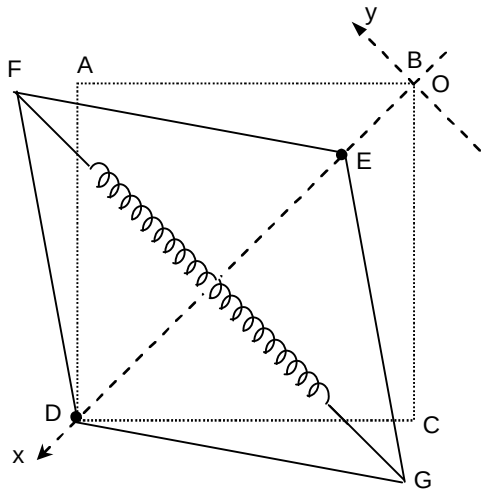
$$\left(\frac{a}{\sqrt{2}} + y\right)^2 + \left(\frac{a}{\sqrt{2}} - \frac{x}{2}\right)^2 = a^2$$

Neglecting x^2 and y^2, we have $y = \frac{x}{2}$

Using COME, we can write

$$\frac{mv^2}{2} + \frac{k}{2}(x)^2 = E = constant$$

$$\Rightarrow T = \pi\sqrt{\frac{4m}{k}}$$



14. Let in time t, m mass of liquid collides with windshield. So

$$F = \left|\frac{p_f - p_i}{t}\right| = \left|\frac{0 - mv}{t}\right| = \frac{mv}{t}$$

$\lambda \Rightarrow$ density of liquid molecules in air so

$m = Svt \times \lambda = Sv\lambda t$

Where $S \Rightarrow$ surface area of the windshield

$F = Sv^2\lambda \Rightarrow$ Mean pressure = $\frac{F}{S} = \lambda v^2$ …(1)

Now as we assumed that λ is the density of liquid molecules of drops in air so in τ time $u\tau S_0$ volume of liquid will strikes on the surface of earth so mass of liquid strikes on the earth in time interval τ will be

$m_0 = S_0 u\tau\lambda$

$\Rightarrow \quad S_0\rho h = S_0 u\tau\lambda$

$\Rightarrow \quad \lambda = \frac{\rho h}{u\tau}$

Putting this value in equation (1), we have

Mean pressure = $\frac{\rho h v^2}{u\tau} = \frac{48}{5} \Rightarrow K = 8$

15. $P_2 = \rho g(h-x),\ P_1 = \rho gh$

$V_2 = Ax \quad V_1 = Ax_0$

Using Boyle's low we can write

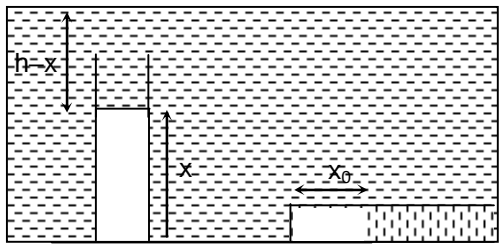
$\rho g(h-x)Ax = \rho gh(Ax_0)$

$\Rightarrow \quad x^2 - hx + hx_0 = 0 \ \Rightarrow\ x = \dfrac{h}{2} \pm \sqrt{\dfrac{h^2}{4} - hx_0}$

$\Rightarrow \quad x = 10m,\ 90m$

$x < \ell$ (length of pipe)

So answer will be x = 10 m



16. $\dfrac{m\ell^2}{12}\alpha = -\left\{\dfrac{2k\theta}{12} + \dfrac{k\theta}{4}\right\}\ell^2 \ \Rightarrow\ \alpha = -\left(\dfrac{5k}{m}\right)\theta$

17. $mg = F_B = \left[\pi R^2 h - \int_0^h \pi\left(R^2 - z^2\right)dz\right]\rho g \quad \Rightarrow m = \dfrac{\pi h^3 \rho}{3}$

$\Rightarrow h^3 = \dfrac{3m}{\pi\rho} = \dfrac{3\times\pi\times 9}{\pi\times 10^3} \ \Rightarrow\ h = \dfrac{3}{10} = 0.3m = 30cm$

18. Mass of water striking vanes/sec $= \rho av$

Initial velocity = v

Final velocity of water = u

Change in velocity = v – u

F = force on the vane $= \rho av(v-u)$

Work done $= f\times u = \rho avu(v-u) = 5400$ watts

$\Rightarrow$ K = 2

19. $\dfrac{h}{d} = \tan\theta$

using triangle EAB

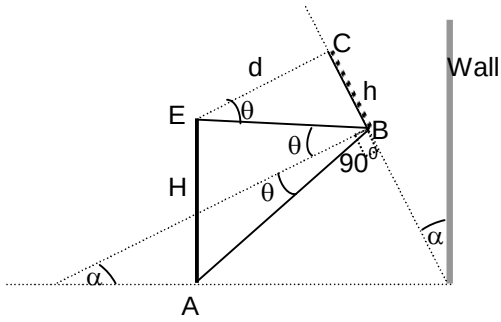
$$\frac{\dfrac{d}{\cos\theta}}{\sin\{90-(\alpha+\theta)\}} = \frac{H}{\sin 2\theta}$$

$$\Rightarrow \quad \frac{d}{\cos(\alpha+\theta)} = \frac{H}{2\sin\theta}$$

$$\Rightarrow \quad 2d = H[\cos\alpha\cot\theta - \sin\alpha]$$

$$= H\cos\alpha\left(\frac{d}{h}\right) - H\sin\alpha$$

$$\Rightarrow \quad h = \frac{Hd\cos\alpha}{2d + H\sin\alpha} = \frac{\frac{3}{2}\times 3\times\frac{3}{5}}{2\times 3 + \frac{3}{2}\times\frac{4}{5}} = \frac{27}{36\times 2} = \frac{3}{8}.$$



20. $\varepsilon_0 VB^2 \equiv \dfrac{\varepsilon_0 E^2}{v^2}V \quad \left[B \equiv \dfrac{E}{v}\right]$

$\equiv \dfrac{u_e}{2v^2}V \equiv \dfrac{\text{Energy}}{(\text{speed})^2} \equiv$ mass, so a = 1, b = 0, c = 0

Chemistry

PART – II

SECTION – A

21. At critical temperature various physical properties of gas and liquid are identical.

22. A. $P = \alpha V$

$PV^{-1} = \alpha$

We know for polytropic process

$PV^x = \text{constant}$

$\therefore x = -1$

We know $C = C_v - \frac{R}{x-1}$

$C = C_v + \frac{R}{2}$

$C = \frac{R}{\gamma - 1} + \frac{R}{2}$

$C = \frac{R(\gamma + 1)}{2(\gamma - 1)}$

B. $dW = -PdV$

$dW = -\alpha V dV$

On integrating above equation.

$W = -\alpha\left[\frac{(4V_o)^2}{2} - \frac{V_o^2}{2}\right]$

$W = -\frac{\alpha}{2}\left[16V_o^2 - V_o^2\right]$

$W = -15V_o^2\frac{\alpha}{2}$

23. $[M(AA'A)_2]$, $[Ma_3b_3]$ and $[M(AB)_3]$ can show fac and mer forms.

24. Say, number of Cu^{2+} ions = x

Number of Cu^{+} ions = 1.8 – x

$\therefore x \times 2 + (1.8 - x) \times 1 = 1 \times 2$

$2x + 1.8 - x = 2$

$x = 2 - 1.8$

$x = 0.2$

% of $Cu^{2+} = \frac{0.2}{1.8} \times 100 = 11.11$

% of Cu^{1+} = 88.89

26. $\Delta G_{mix} = RT\left(n_A \ell n \chi_A + n_B \ell n \chi_B\right)$

$= 2 \times 300\left(1\,\ell n\frac{1}{3} + 2\,\ell n\frac{2}{3}\right)$

$\Delta G_{mix} = 600\left(-1.0986 - 0.8109\right) = -1145.7$ cal

$\Delta S_{mix} = -\frac{\Delta G_{mix}}{T} = 3.82$ cal k^{-1}

$\Delta H_{mix} = 0$

28. In froth floatation process pine oil used as collectors which enhance non-wettability of mineral particles.
- Cresol is froth stabilizers.
- NaCN used as depressant

29. $C + 4HNO_3 \longrightarrow CO_2 + 4NO_2 + 2H_2O$

$Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

$Zn + 4HNO_3 \longrightarrow Zn(NO_3)_2 + 2NO_2 + 2H_2O$

$Sn + 4HNO_3 \longrightarrow H_2SnO_3 + 4NO_2 + H_2O$

30. (A) $H_2O + Cl_2 \longrightarrow 2HCl + O$ (B) $H_2S + Cl_2 \longrightarrow 2HCl + S$

(C) $C_{10}H_{16} + 8Cl_2 \longrightarrow 16HCl + 10C$ (D) $8NH_3 + 3Cl_2 \longrightarrow 6NH_4Cl + N_2$

SECTION-B

31. Pb^{2+}: $Pb^{2+} + 2KI \longrightarrow \underset{\text{Yellow}}{PbI_2}\downarrow + 2K^+$

$Pb^{2+} + K_2CrO_4 \longrightarrow \underset{\text{Yellow}}{PbCrO_4}\downarrow + 2K^+$

Ag^+: $2Ag^+ + 2NaOH \longrightarrow \underset{\text{Brown}}{Ag_2O}\downarrow + H_2O + 2Na^+$

$2Ag^+ + H_2S \longrightarrow \underset{\text{Black}}{Ag_2S}\downarrow + 2H^+$

Hg_2^{2+}: $Hg_2^{2+} + 2NaOH \longrightarrow \underset{\text{Black}}{Hg_2O}\downarrow + H_2O + 2Na^+$

$Hg_2^{2+} + H_2S \longrightarrow \underbrace{Hg\downarrow + HgS\downarrow}_{\text{Black}} + 2H^+$

$Hg_2^{2+} + 2KI \longrightarrow Hg_2I_2 + 2K^+$

$Hg_2I_2 + 2KI \longrightarrow K_2[HgI_4] + \underset{\text{Black}}{Hg}\downarrow$

32. (A). Process is isobaric

$C_V = \frac{5R}{2}$ $C_P = \frac{7R}{2}$

$|q_p| = nC_P\Delta T = \frac{7RT_o}{2}$

$|\Delta U| = nC_V\Delta T = \frac{5RT_o}{2}$

$|W| = RT_o$

(B). Process is isochoric, $C_V = \frac{3R}{2}$

$|q_V| = |\Delta U| = \frac{3RT_o}{2}$

$|W| = 0$

(C). $C_V = \frac{R}{\gamma - 1} = \frac{R}{1.5 - 1} = 2R$

$VT = \text{constant}$ $PV^2 = \text{constant}$ $\frac{P}{\rho^2} = \text{constant}$

$V + T\dfrac{dV}{dT} = 0 \qquad \dfrac{dV}{dT} = -\dfrac{V}{T} \qquad \dfrac{P}{n}\dfrac{dV}{dT} = -\dfrac{PV}{nT} = -R$

$|W| = RT_o \qquad |\Delta U| = 2RT_o \qquad |q| = RT_o$

(D). $\gamma = 1 + \dfrac{2}{4} = \dfrac{3}{2}$

$C_V = \dfrac{R}{\gamma - 1} = 2R$

PT = constant $\quad P^2V$ = constant

$\dfrac{P^2}{\rho} = \text{constant}$

$|\Delta U| = 2RT_o$

SECTION – C

33. Ionisation isomerism
Linkage isomerism
Geometrical isomerism

34. $\underset{n_f=2}{3ZnSO_4} + 2K_4\underset{n_f=3}{\left[Fe(CN)_6\right]} \longrightarrow K_2Zn_3\left[Fe(CN)_6\right]_2 + 3K_2SO_4$

Milliequivalents of $ZnSO_4$ = milliequivalents of $K_4[Fe(CN)_6]$

$N_1V_1 \quad = \quad N_2V_2$

Volume of $K_4[Fe(CN)_6]$ **(V_2)** $= \dfrac{60 \times 0.01 \times 2}{0.05 \times 3} = 8$ ml

35. $\Delta T_f = K_f m$

$m = \dfrac{0.93}{1.86} = 0.5$

$\Delta T_b = K_b m$

$K_b = \dfrac{0.26}{0.5} = 0.52\,K\ kg\,mol^{-1}$

Similarly $\Delta T_b' = K_b m'$

$\Delta T_b' = \dfrac{K_b \times w \times 1000}{W \times m.wt}$

$\therefore$ m.wt. of solute $= \dfrac{K_b \times w \times 1000}{W \times \Delta T_b'}$

$= \dfrac{0.52 \times 7.87 \times 1000}{100 \times 0.44}$

$31x \simeq 93$

$x = 3$

36. $2H_2 + O_2 \longrightarrow 2H_2O$

$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

$2CO + O_2 \longrightarrow 2CO_2$

$C_2H_2 + \dfrac{5}{2}O_2 \longrightarrow 2CO_2 + H_2O$

Required oxygen for the combustion of given gas = 1.04 m^3

Hence, volume of air required $= \frac{1.04 \times 100}{20.8} = 5\ m^3$

37. Al, Cr, Fe

38. Telluric acid H_6TeO_6 is quite different from sulphuric and selenic acid. It exists as octahedral $Te(OH)_6$ molecule in the solid. It is a weak dibasic acid and forms two series of salt, $NaTeO(OH)_5$ and $Na_2TeO_2(OH)_4$.

39. $Zn(s) + 2Ag^+(aq) \rightleftharpoons Zn^{2+}(aq) + 2Ag(s)$

$E^o_{cell} = \frac{0.059}{2} \log_{10} K_{eq}$

$0.8 + 0.76 = \frac{0.059}{2} \log_{10} K_{eq}$

$K_{eq} = 7.61 \times 10^{52}$

As K_{eq} is very large the reaction will tend towards completion

Moles of zinc added $\frac{6.53 \times 10^{-2}}{65.3} = 10^{-3}$

Moles of Ag^+ in solution $= \frac{10^{-5} \times 100}{1000} = 10^{-6}$

As Ag^+ is limiting reagent, the moles of Ag precipitated $\simeq 10^{-6}$

Hence x = 6

40. $P_1V_1 = P_2V_2$

P_2 = pressure of H_2 gas $= \frac{60 \times 10}{20}$

P_2 = 30 cm of Hg

Final total pressure over liquid water (P) = 30 + 20 = 50 cm of Hg

$\frac{P}{10} = \frac{50}{10} = 5$

Mathematics

PART – III

SECTION – A

41. D_m will be open finally if m is the perfect square

42. (A) $(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + ... + {}^nC_nx^n$

Put $x = \omega$ where $\omega = -\frac{1}{2} + \frac{i\sqrt{3}}{2}$

$$\left({}^nC_0 + {}^nC_3 + {}^nC_6 + ...\right) + ... + \left({}^nC_1 + {}^nC_4 + {}^nC_7 + ...\right)\omega + \left({}^nC_2 + {}^nC_5 + {}^nC_8 + ...\right)\omega^2 = (1+\omega)^n$$

$$\left({}^nC_0 + {}^nC_3 + {}^nC_6 + ...\right) + \left({}^nC_1 + {}^nC_4 + {}^nC_7 + ...\right)\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) +$$

$$\left({}^nC_2 + {}^nC_5 + {}^nC_8 + ...\right)\left(-\frac{1}{2} - \frac{i\sqrt{3}}{2}\right) = \left(-\omega^2\right)^n$$

$$\left({}^nC_0 + {}^nC_3 + {}^nC_6 + ...\right) - \frac{1}{2}\left({}^nC_1 + {}^nC_2 + {}^nC_4 + {}^nC_5 + ...\right) +$$

$$\frac{i\sqrt{3}}{2}\left({}^nC_1 - {}^nC_2 + {}^nC_4 - {}^nC_5 + ...\right) = \left(-\omega^2\right)^n$$

Take modulus both sides

(B) $a^5b^2 = 4 \Rightarrow 5\log_2(a) + 2\log_2(b) = 2$

$y = 10\log_2 a \cdot 4\log_2 b = 40\log_2 a \cdot \log_2 b$

A.M. $\geq$ G.M. $\Rightarrow \frac{5\log_2(a) + 2\log_2(b)}{2} \geq \sqrt{10\log_2 a \cdot \log_2 b}$

$\log_2 a \cdot \log_2 b \leq \frac{1}{10} \Rightarrow y \leq 4$, $y = 4$ when $5\log_2 a = 2\log_2 b \Rightarrow a^5 = b^2 = 2$

(C) For constant term put $x = 0$ and get constant term = 4

(D) Coefficient of $x^{24} = 1.25 + 2.24 + 3.23 + ... + 25.1 = 2925$

43. Roots of the equation $y^3 - x_3y^2 - x_2y - x_1 = 0$ are 3, 5 and 7

$x_3 = 15$, $x_2 = -71$, $x_1 = 105$

44. Let R be the common ratio of the G.P. and D be the common difference of A.P.

$a_5 = a_5$, $a_9 = Ra_5$, $a_{16} = a_5R^2$

$a_9 - a_5 = 4D \Rightarrow (R - 1)a_5 = 4D$ ..... (1)

$a_{16} - a_9 = 7D \Rightarrow R(R - 1)a_5 = 7D$ ..... (2)

From equation (1)/(2), we get $\frac{1}{R} = \frac{4}{7} \Rightarrow R = \frac{7}{4}$

From equation (2) – (1), we get $(R - 1)^2a_5 = 3D \Rightarrow \frac{9a_5}{16} = 3D$

$$\frac{3}{16}(a_1 + 4D) = D \Rightarrow \frac{3}{16}a_1 = \left(1 - \frac{3}{4}\right)D \Rightarrow a_1 = \frac{4D}{3}$$

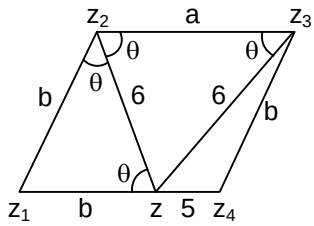
45. $a = b + 5$

and $\frac{b}{6} = \frac{6}{a}$

$\Rightarrow ab = 36$

$b^2 + 5b - 36 = 0$

$\Rightarrow b = 4$



46. $\left(\frac{n}{3}\right)^n < n!$ put $n = 300 \Rightarrow 100^{300} < 300!$ ..... (1)

$3 < 100 \Rightarrow 3^{150} < 100^{150} \Rightarrow (300)^{150} < 100^{300}$

$\sqrt{300^{300}} < 100^{300}$ ..... (2)

From equation (1) and (2), we get $\sqrt{300^{300}} < 300!$

47. (A) $a_{ij} = -a_{ji} \Rightarrow a_{ij} + a_{ji} = 0 \Rightarrow$ A is skew symmetric matrix

(C) $A^2 = 2A \Rightarrow A^3 = 2^2A \Rightarrow A^6 = 2^5A$

(D) A^6B^7 is a skew symmetric matrix of odd order

48. CD : $\vec{r} = \left(\hat{i} - 2\hat{j} + 4\hat{k}\right) + \frac{\lambda}{3}\left(7\hat{j} - 7\hat{k}\right)$

BE : $\vec{r} = \left(-\hat{i} + \hat{j} + \hat{k}\right) + \frac{\mu}{3}\left(7\hat{i} - 7\hat{j} + 7\hat{k}\right)$

$P \equiv \left(\hat{i} - \hat{j} + 3\hat{k}\right)$

Area of tetrahedron ABCF

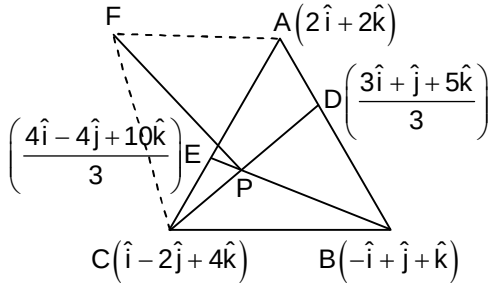
$= \frac{1}{3}(\text{Area of base triangle}) \times \text{height} = \frac{7}{3}$ cubic units

$\overrightarrow{AB} \times \overrightarrow{AC} = 7\hat{j} + 7\hat{k}$, $\left|\overrightarrow{PF}\right| = PF = \sqrt{2}$ units

$\overrightarrow{PF} = \sqrt{2}\left(\frac{7\hat{j} + 7\hat{k}}{\sqrt{49 + 49}}\right) = \hat{j} + \hat{k}$ = P.V. of F – P.V. of P

P.V. of F = $\hat{i} + 4\hat{k}$

Vector equation of AF is $\vec{r} = 2\left(\hat{i} + \hat{k}\right) + \alpha\left(-\hat{i} + 2\hat{k}\right)$

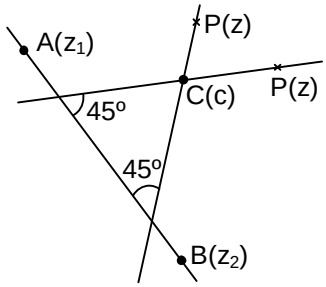


49. By eliminating n $2\left(\frac{\ell}{m}\right)^2 - \frac{\ell}{m} - 1 = 0$

$\Rightarrow \frac{\ell_1}{m_1} = 1$, $\frac{\ell_2}{m_2} = -\frac{1}{2}$

On solving, we get $(\ell_1, m_1, n_1) = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\sqrt{\frac{2}{3}}\right)$ and $(\ell_2, m_2, n_2) = \left(\frac{1}{\sqrt{6}}, -\sqrt{\frac{2}{3}}, \frac{1}{\sqrt{6}}\right)$

50. Let P(z) be any point on the required line.



Then, $\frac{\overrightarrow{CP}}{\left|\overrightarrow{CP}\right|}$ i.e. $\frac{z-c}{|z-c|}$ is a unit vector parallel to it

Let $A(z_1)$ and $B(z_2)$ be two points on

$\bar{a}z + a\bar{z} + b = 0$ then $\frac{z_2 - z_1}{|z_2 - z_1|}$ is a unit vector parallel to the line

$a\bar{z} + \bar{a}z + b = 0$

$$\frac{z-c}{|z-c|} = \frac{z_2 - z_1}{|z_2 - z_1|} e^{\pm i\left(\frac{\pi}{4}\right)}$$

$$\frac{(z-c)^2}{(z-c)(\bar{z}-\bar{c})} = \frac{(z_2 - z_1)^2}{(z_2 - z_1)(\bar{z}_2 - \bar{z}_1)} e^{\pm i\frac{\pi}{2}}$$

$$\frac{z-c}{\bar{z}-\bar{c}} = \pm i\left(\frac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1}\right) \quad \text{..... (1)}$$

$\because$ $A(z_1)$ and $B(z_2)$ are on the line $a\bar{z} + \bar{a}z + b = 0$ therefore $a\bar{z}_1 + \bar{a}z_1 + b = 0$

$a\bar{z}_2 + \bar{a}z + b = 0$

$$\Rightarrow -\frac{a}{\bar{a}} = \frac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1} \quad \text{..... (2)}$$

From equation (1) and (2), we get $\frac{z-c}{a} \pm \frac{i(\bar{z}-\bar{c})}{\bar{a}} = 0$

SECTION – B

51. (A) $f(x) = x^3 + ax^2 + bx + c = (x^2 + 1)(x + a) + (b - 1)x + (c - a)$
$\because$ $f(x)$ is divisible by $x^2 + 1 \Rightarrow b - 1 = 0 \Rightarrow c = a$

(B) $(A + B)(A - B) = (A - B)(A + B) \Rightarrow AB = BA$, $A^T = A$, $B^T = -B$
$(AB)^T = (-1)^{k-1}AB \Rightarrow k$ is an odd number

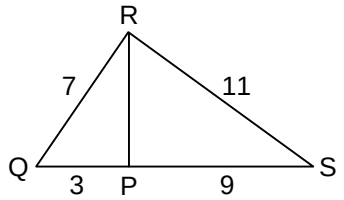
(C) $10^{100} - 43 = 999$ 9957
Sum of digits = 98(9) + 5 + 7 = 894

(D) $x = 7\cos\theta - \sin\theta$, $y = 7\sin\theta + \cos\theta$, $z = 10$
$\frac{x^2 + y^2}{z} = 5$

52. (A) $(x + y^2)(x - y^2) = 17^2$
$x + y^2 = 17^2$ and $x - y^2 = 1$ or $x + y^2 = 17$ and $x - y^2 = 17$
or $x + y^2 = -1$ and $x - y^2 = -17^2$ or $x + y^2 = -17$ and $x - y^2 = -17$
Possible cases: $(x, y) \rightarrow (\pm 145, \pm 12), (\pm 17, 0)$
$x + 12y + 4 = 293, 5, 21, -13, 3, -285$

(B) $PQ^2 + RS^2 = QR^2 + SP^2 = 130$

$$\frac{\overrightarrow{PS} \cdot \overrightarrow{QR}}{9} = \frac{9 \times 7 \times \frac{3}{7}}{9} = 3$$



(C) $x + y + z = 20$, number of the solution is ${}^{19}C_2$ they form triangle if $1 \le x, y, z \le 9$
Number of solutions = coefficient of x^{20} in $(x^1 + x^2 + ... + x^9)^3 = {}^{19}C_2 - 3^{10}C_2$

Required probability $P = \frac{{}^{19}C_2 - 3 \times {}^{10}C_2}{{}^{19}C_2} \Rightarrow P = \frac{4}{19}$

(D) Let the series be 21, 21r, $21r^2$, $21r^3$, ...

Sum = $\frac{21}{1-r} \in N$ and $21r \in N$

$\frac{21 \times 21}{21-21r} \in N \Rightarrow 21 - 21r \in N$ and $21r < 21$

$21 - 21r$ may be equal to 1, 3, 7, 9

$21r = 20, 18, 14, 12$

$|k - 15| = 5, 3, 1$

SECTION – C

53. $(x^4 + 2x^3 + bx^2 + cx + d) = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3)(x - \alpha_4) = (x^2 + px + q)(x^2 + px + r)$
$= x^4 + 2px^3 + (q + r + p^2)x^2 + (pr + pq)x + qr$
$(2p)^3 - 4(2p)(q + r + p^2) + 8(pq + pr) = 0$
$2^3 - 4 \times 2 \times b + 8c = 0$
$\Rightarrow b - c = 1$

54. Probability that B wins the match in the 4th game
$= \frac{1}{2} \times \frac{1}{6} \times \frac{1}{3} \times \frac{1}{2} \times 6 + {}^3C_1 \times \frac{1}{2} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{2} = \frac{6}{72} + \frac{3}{36} = \frac{6+6}{72} = \frac{1}{6}$

55. $x^3 + 3x^2 + k = 0 \rightarrow \alpha, \beta, \gamma$
$\alpha + \beta + \gamma = -3$, $\alpha\beta + \beta\gamma + \gamma\alpha = 0$, $\alpha\beta\gamma = -k$
$\alpha^2 + \beta^2 + \gamma^2 = 9$
$\because \alpha, \beta, \gamma \in I \Rightarrow \alpha^2 = 9, \beta^2 = 0, \gamma^2 = 0$
or $\alpha^2 = 4, \beta^2 = 4, \gamma^2 = 1$
$\alpha^2 = 4, \beta^2 = 4, \gamma^2 = 1$
Possible roots: $\pm 3, 0, 0, \pm 2, \pm 2, \pm 1$
But $\alpha\beta + \beta\gamma + \gamma\alpha = 0$
So, possible roots are 3, 0, 0; –3, 0, 0; 2, 2, –1; –2, –2, 1
Possible non-zero values of k are –4 and 4

56. $A_k \cdot A_{k+1} = \begin{bmatrix} a_{k-1} & 0 \\ 0 & a_k \end{bmatrix} \begin{bmatrix} a_k & 0 \\ 0 & a_{k+1} \end{bmatrix}$

$A_k \cdot A_{k+1} = \begin{bmatrix} a_{k-1} \cdot a_k & 0 \\ 0 & a_k a_{k+1} \end{bmatrix}$

$A_k \cdot A_{k+1} = \begin{bmatrix} {}^nC_{k-1}\ {}^nC_k & 0 \\ 0 & {}^nC_k\ {}^nC_{k+1} \end{bmatrix}$

$B = \begin{bmatrix} \sum_{k=1}^{n-1} {}^nC_{k-1} \cdot {}^nC_k & 0 \\ 0 & \sum_{k=1}^{n-1} {}^nC_k \cdot {}^nC_{k+1} \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$

$a = b = {}^{2n}C_{n-1} - n$

57. $(r^2 + 2)(r + 1)! + 2r(r + 1)! = (r + 1)(r + 2)! - r(r + 1)!$

58. $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)$
$a + b + c = e^x$, $a + b\omega + c\omega^2 = e^{\omega x}$, $a + b\omega^2 + c\omega = e^{\omega^2 x}$

59. det(A) is twice of area of triangle with vertices (ℓ, p), (m, q), (n, r) with sides 3, 4, 5

$\Delta^2 = s(s-a)(s-b)(s-c)$
$\Rightarrow \Delta^2 = 6 \times 3 \times 2 \times 1$
$\Rightarrow \Delta = 6$
Now det(A) = 12
$\Rightarrow$ det(B) = 144

60. $a + ib = z \Rightarrow |z|^{5015} - |z|^3 = 0$
$|z|^3 (|z|^{5012} - 1) = 0$
$|z| = 0$ or $|z|^{5012} = 1 \Rightarrow |z| = 1 \Rightarrow z\bar{z} = 1$
$z = 0$

$z^{5015} = \bar{z}^3 \Rightarrow z^{5015} = \frac{1}{z^3} \Rightarrow z^{5018} = 1$

Number of complex numbers = 5018 + 1 = 5019